[METRIC]

A-A-2767
October 2, 1992

COMMERCIAL ITEM DESCRIPTION

INK, DRAWING, WATERPROOF, BLACK

This Commercial Item Description, prepared by the General Services Administration, Office Supplies & Paper Products Commodity Center, New York, NY 10278, is approved for promulgation. This document supersedes Federal Specification TT-I-528, which is canceled.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers black, waterproof drawing ink for general drafting use, drawing, and tracing.

1.2 Classification.

1.2.1 Types. The ink shall be of the following types, as specified (See section 6).

Type I.- With temperate and tropical temperature-resistant characteristics.

Type II.- With low temperature-resistant characteristics.

1.2.2 Sizes (bottles of ink). (See section 5)

1.2.3 Colors of ink. The ink shall be black or as specified (see section 6):

2. APPLICABLE DOCUMENTS.

The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

FSC 7510

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2.1 Public Law and Code of Federal Regulations:

Code of Federal Regulations and Federal Acquisition Regulations are for sale by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

Public Law 94-580 - Use of Recovered Materials

16 CFR Part 1500 - Federal Hazardous Substances Labeling Act.

2.2 National Motor Freight Traffic Association, Inc. Agent:

National Motor Freight Classification.

Application for copies shall be addressed to the American Trucking Associations, Inc., 2200 Mil Rd., Alexandria, VA 22314.

2.3 Uniform Classification Committee Agent:

Uniform Freight Classification.

Application for copies shall be addressed to the National Railroad Freight Committee, Suite 1120, 222 South Riverside Plaza, Chicago, Illinois 60606.

2.4 Federal Standards, Specifications, and Commercial Item Descriptions:

Federal standards and specifications are available from GSA Business Service Centers and GSA Specification Unit (3FBP-W), 7th and D streets SW., Washington, D.C. 20407.

Single copies of this specification and other specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Auburn, Washington.

FED-STD-123 Marking for Domestic Shipment (Civil Agencies)

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

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2.5 Military Standards:

Military Standards are available from the Standardization Documents Order Desk; Bldg. 4D; 700 Robbins Avenue; Philadelphia, PA 19111-5094.

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 Marking for Shipment and Storage.

3. REQUIREMENTS

3.1 Composition and stability of ink. The ink shall be composed of finely divided carbon in a stable state of suspension in a liquid medium. The ink shall not show any coagulation, granulation, or precipitation under working and storage conditions (see 6.3). Evidence of instability shall not be present when the ink is received nor when the sample ink is subjected to the designated stability tests specified in 4.4.1.

3.2 Performance. The ink as received and after exposure as specified in 4.4.1 shall have the following performance characteristics when tested as specified in 4.4.2: During the test, the ink shall not dry in the pen. The ink shall flow freely and produce lines of uniform width on the specified drawing paper. The color shall be intense black and the ink shall have a high degree of opacity and a low degree of gloss. The ink shall show no signs of abrasion when the test marking is cleaned with an eraser of the "art-gum" type as specified in 4.4.2.4.

3.3 Resistance to solvents. The ink, when thoroughly dried, shall be insoluble in cold water and in organic solvents, when tested as specified in 4.4.3.1 and 4.4.3.2.

3.4 Specific gravity. The ink shall have a specific gravity at 25°C. of not less than 1.055 when determined as specified in 4.4.4.

3.5 Identification marking. Each bottle of ink shall be clearly and legibly lithographed or marked with water-resistant ink, or furnished with a label printed with water-resistant ink on a good grade of printing paper and firmly affixed to the bottle with a water-insoluble adhesive. The markings on the bottle and its individual box shall clearly identify the product and its manufacturer in accordance with standard commercial practice. (See section 5)

3.6 Workmanship The ink shall be a smooth, homogeneous, black liquid free from lumps, string, and objectionable odor. Labels shall be neatly affixed and clearly printed. Ink containers shall show no evidence of spilling or leakage. The ink shall be free of defects which affect appearance or usability.

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3.7 Hazardous Materials.

3.7.1 Toxicity - Use of any toxic substance must be in accordance with all applicable federal statutes. The harmful substances shall be defined as the poison and injurious substances as designated in Fed Std. 313. The vendor shall prepare a certification that the inks conforms to the above requirements.

3.7.2 Material safety data sheets - Safety data shall be furnished in accordance with Fed. Std. 313. The pertinent government mailing addresses for submission of data are listed in appendix B of Fed. Std. 313.

3.7.3 Labeling - Containers shall be labeled to comply with the "Federal Hazardous Substances Act Regulations", 16 Code of Federal Regulations, Part 1500.

3.8 Shelf life. The markers shall be able to meet all the requirements of this specification from the time of delivery to a minimum of 12 months in storage. Suppliers shall replace, without cost to the Government, all markers failing to meet this requirement.

4. QUALITY ASSURANCE REQUIREMENTS.

4.1. Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2. Sampling and inspection. Lot definition - Each lot of markers shall be inspected for conformance to the requirements herein. Inspection shall be in accordance with MIL-STD-105 using the sampling plans specified below. The lot shall be expressed in markers manufactured from the same materials, at essentially the same time.

4.3 Inspection.

4.3.1 Inspection of materials and components. The supplier is responsible, in accordance with 4.1, above, for insuring that components and materials used were manufactured, tested, and inspected in accordance with the specified requirements of referenced subsidiary specifications and standards and in accordance with this specification, as applicable.

4.3.1.1 Examination of the end item.

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4.3.1.1.1 Visual defects - The sample unit for this examination shall be one bottle of ink. A random sample shall be taken from each lot in accordance with Inspection Level S-3. The AQL shall be 2.5 percent defective. Samples shall be examined for defects listed below. Any bottle of ink containing one or more defects shall be rejected.

Examine	Defects
Workmanship	See Workmanship paragraph
Material	Not as specified
Appearance	Not homogeneous. Contains lumps or strings. Stray spilling of ink or leakage on outside of bottle.
Design	Bottle shape or closure not as specified, or not commonly used for inks.
Fill of individual bottle	Empty bottle, or overfilling to extent that bottle may burst or leak in hot weather due to lack of liquid expansion space.

4.3.1.1.2 Dimensional defects - The sample unit for this examination shall be one bottle of ink. A random sample shall be taken from each lot in accordance with Inspection Level S-2. The AQL shall be 4.0 percent defective. Samples shall be examined for defects listed below. Any bottle of ink containing one or more defects shall be rejected.

Examine	Defects
Size of bottle	Not as specified
Quantity in bottle	Not as specified

4.3.1.2.3 Sampling of end items for performance tests - Sampling for performance testing shall be performed in accordance with MIL-STD-105. A random sample from each lot in accordance with Inspection Level shall be S-2 with an AQL of 4.0 percent defective. The sample unit shall be one bottle of ink.

4.3.1.2.4 Packaging, Packing, and Marking - Inspection Level S-2 with an AQL of 4.0 percent defective. The sample unit shall be one fully prepared shipping container inspected for conformance with the requirements of section 6.

4.3.2.1 Testing of ink. Each sample unit shall meet all requirements. One sample unit shall consist of 1 pint from one or more bottles kept at ordinary room temperature. Another sample unit of 1 pint shall consist of ink from one or more bottles after exposure is made in the bottles as specified in 4.4.1.2.

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4.4 Tests.

4.4.1 Stability of ink.

4.4.1.1 Original stability. Examine the sample of ink in its original container for precipitation by inserting a glass rod. If present, invert bottle and shake vigorously for one minute. Examine again for sediment and coagulation. Place a few drops of ink between glass plates and rub plates together and examine for granulation. (See 6.3.)

4.4.1.2 Stability to temperature changes. Samples of ink in their original closed containers shall be exposed, one container to both of the temperature conditions for the applicable type, as follows:

Type I: (a) At 51° +/- 2° C. for four hours.
(b) At 0° +/- 2° C. for four hours.

Type II: (a) At 71°+/- 2°C. for. four hours.
(b) At -53 +/- 2°C. for three days.

The inks after exposure to the above specified temperatures shall be examined for coagulation, granulation, tested as specified in 4.4.1.1.

4.4.2 Performance. The performance tests shall be made with clean and properly sharpened drawing pens and the ink shall be observed for drying out in the pens, during the tests.

4.4.2.1 Ink flow.

4.4.2.1.1 Test 1. Set a drawing pen at an opening of 76 microns +/- 25 microns (0.003 +/- 0.001 inches) and fill it with sufficient ink to draw at least five lines, each 18 cm long. The lines shall be drawn on drawing paper The ink shall be considered unsatisfactory if it is not possible to draw five clean, sharp, unbroken lines of uniform width.

4.4.2.1.2 Test 2. Set a detail drawing pen with at, opening of 635 microns +/- 25 microns (0.025 +/- 0.001 inches) and draw 10 lines 5 cm long, on both plain paper. After the lines are allowed to dry, lines of the same width and length shall be drawn at right angles across each of the sets previously drawn. the intersecting lines shall be free from distortion or blots.

4.4.2.2 Penetration. A sample of the markings prepared in 4.4.2.1.2 shall be rubbed with an eraser of the "Van Dyke" No. 6500 type or equal. The ink markings shall be completely erasable.

4.4.2.3 Color, opacity ,and gloss. Draw two lines, 30 cm long on plain paper. After the lines are dry, apply another coat on the

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second line with the second ink. Examine the middle portion of each line for intensity of color (blackness), opacity, and degree of gloss.

4.4.2.4 Abrasion resistance. A sample of marking, prepared as specified in 4.4.2.3, shall be rubbed vigorously with, an eraser of the "art-gum" type or equal. There shall be no perceptible evidence of ink removal.

4.4.3 Fastness tests. Draw two parallel Lines, approximately 635 microns (0.025 inch wide), 2.5 cm apart, 20 cm long on drawing paper, and mark off into two equal parts. After these have been allowed to dry for at least 1 hour, each part shall he subjected to the following tests:

4.4.3.1 Solubility in water. Place four drops of water 21°-26°C. (70°-80°F.) and spread it over the lines. Allow the water to evaporate naturally. There shall be spreading of the lines.

4.4.3.2 Resistance to organic solvents. Repeat the above operation using hexane on separate areas. Additional quantities should be added in order to keep the surface wet for at least 5 minutes. Observe for evidence of dissolution of the ink, spreading of the lines, or reduction of intensity of color (blackness).

4.4.4 Specific gravity. The specific gravity of ink shall be determined at 25° C. by using a pycnometer.

5. PREPARATION FOR DELIVERY

5.1. Unless otherwise specified (see section 6), preparation for delivery shall conform to normal commercial practice.

5.1.1 Unit package.

5.1.1.1 Type I. Three-quarters, one, two, four, six, eight, sixteen, and thirty-two ounces of ink shall be packaged in 22, 29.6, 59.1, 118, 177, 236, 473, 946 ml (3/4, 1-, 2-, 4-, 6-, 8-, 16-, and 32-ounce) bottles, respectively, and shall be furnished as specified (see 6.2). The bottles shall be fabricated from commercial grade glass or polyethylene material. Each 22 ml (3/4 ounce) and 29.6 ml (1-ounce) bottle shall be provided with a leak proof stopper and ink applicator in accordance with standard commercial practice. The stopper and applicator shall not be affected by the ink, nor shall the ink be affected by the stopper and applicator.

5.1.1.2 Type II. Type II ink shall be packaged 37 ml (1 1/4 ounces) per bottle. Each bottle shall be molded in one piece from polyethylene plastic and shall be of the flexible squeeze-bottle type fitted with an elongated polyethylene plastic or urea formaldehyde plastic applicator tip which will facilitate the dispensing of one drop of ink at a time to drafting and lettering

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pens and which will prevent ink from spilling from the bottle when it is accidentally overturned with the cap removed. A screw-type urea formaldehyde plastic cap which screws to the bottle and effectively seals the applicator tip when in place, shall be provided. Each bottle and cap shall be sufficiently rugged to withstand field handling and use without damage.

5.2. All packages shall be marked to show month and year of manufacture (date of marker assembly).

5.2.1. Packaging, and packing - Packaging, and packing shall be as indicated below, unless specified otherwise in the contract or purchase order (see section 6). The normal commercial number of unit packages shall be packed in a close-fitting fiberboard box used commercially for this item. The complete pack shall be in compliance with the Uniform Freight Classification and the National Motor Freight Classification. The weight of the shipping container shall not exceed 30 Kg.

5.2.2. Marking - Marking shall be as indicated below, unless specified otherwise in the contract or purchase order (see section 6).

5.2.2.1. For Shipment To Civil agencies - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123. Bar code marking is required, unless specified otherwise (See section 6).

5.2.2.2. For Shipment To Military activities - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Bar code marking is required, unless specified otherwise (See section 6).

5.2.2.3. Additional marking - The unit and shipping containers shall include the date of manufacture and the date of reinspection (12 months after the date of manufacture) and the shipping containers, in addition, shall include any precautionary marking required by the National Motor Freight Classification Item 222, Uniform Freight Classification Rule 41, and Department of Transportation regulations.

6. NOTES

6.1 Intended use.

6.1.1 Type I drawing ink covered herein is intended for use in tropic and temperate climates only.

6.1.2 Type II is intended for use under conditions where ink may be subjected to extremely low temperatures.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this commercial item description.
(b) Purchasers shall indicate color of ink required (see 1.2.1).
(c) Purchasers shall specify arrangement for inspection and inspection facilities, if other than specified.
(d) Purchasers shall specify packaging, packing, and any special marking for shipment, if other than required (See section 5).
(e) Identification marking, if other than required.
(f) If bar code marking is not required.
(g) MSDS's required, in addition to those required by Fed-Std-313.
(h) Purchasers should specify palletization , if required.
(i) Type required (see section 1).
(j) Size of unit package for type I ink (see section 5).

6.3 Definitions. The following terminology included in this specification is defined as follows:

(a) Coagulation - The result of gelation.

(b) Granulation - Hard particles which can be detected by placing a few milliliters of ink between glass plates and then rubbing the plates together.

(c) Precipitation - Precipitates are defined as particles of solids which fall out of solution or suspension and which cannot be re-dispersed or suspended by either inverting or shaking bottle by hand for 1 minute.

MILITARY INTERESTS:

NONE: DOD has determined that no military activity has an official interest in this commercial item description.

Preparing Activity:
GSA-FSS

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